

CDFD Runtime Paper 03: Dynamics, Regimes, and Stability Tests

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Abstract

Executive synthesis. This paper states the dynamical core of CDFD as an executable model. Flow Φ diffuses through adaptive constraint C ; the medium changes through stress response, relaxation, and diffusion; and the surface-memory product SM_s can stabilize or destabilize the operating ratio $\Psi_s = (\Phi/C)SM_s$. The local experiments show why bounded numerical diagnostics matter: the same equations that produce recovery under moderate stress can produce runaway hyper-adaptation when recovery is too weak.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Master Flow Equation

CDFD begins with the assumption that effective conductivity is inversely related to constraint. In a coarse spatial model,

$$\frac{\partial \Phi}{\partial t} = J_{\text{in}} + \nabla \cdot \left(\frac{1}{C} \nabla \Phi \right) + S_{\text{src}} - D_{\text{sink}}, \quad (1)$$

where J_{in} is an optional external input, S_{src} is a source term, and D_{sink} is a dissipation or removal term.

3 Adaptive Constraint Equation

The constraint field evolves according to the stress induced by changing flow:

$$\frac{\partial C}{\partial t} = \alpha \left| \frac{\partial \Phi}{\partial t} \right| - \beta C + \gamma \nabla^2 C. \quad (2)$$

Here α is stress-driven constraint accumulation, β is recovery or forgetting, and γ is spatial spreading of constraint. These parameters are model parameters, not universal constants.

4 Surface and Memory Dynamics

The current engine includes an adaptive surface channel:

$$\frac{\partial S}{\partial t} = a\Phi - bC + cM_s - dS + \xi \nabla^2 S. \quad (3)$$

The term $\xi \nabla^2 S$ matters because it lets neighboring cells buffer local surface amplification. Without sufficient recovery or diffusion, S can become an amplifier rather than a stabilizer.

The structural memory channel is modeled as a hysteresis process:

$$M_s(t + \Delta t) = 1 + (M_s(t) - 1)e^{-\Delta t/\tau_M} + \mu H_{\text{overload}} + \lambda_M \nabla^2 M_s, \quad (4)$$

where H_{overload} is a bounded overload contribution. This is the mechanism tested by the vacuum-hysteresis experiment.

5 Regime Classification

The runtime uses the operating ratio

$$\Psi_s = \frac{\Phi}{C} S M_s \quad (5)$$

as a local regime variable. Typical labels are:

- $\Psi_s < 0.8$: constrained or under-driven;
- $0.8 \leq \Psi_s \leq 1.2$: balanced operating band;
- $\Psi_s > 1.2$: overload or stress regime;
- sustained high Ψ_s : rupture, collapse, escape, or rerouting, depending on domain mapping.

6 Experiment Links

6.1 Hyper-adaptation boundary

`experiments/notebooks/discovery_adaptive_surface.py` and the release selection report identify an overloaded run in which Ψ_s moves from approximately 5.15 to approximately 979. This output is not a successful engineering result. It is an instability boundary: a test case showing that high Φ , low C , high S , high M_s , and weak recovery can produce runaway amplification.

6.2 Vacuum-memory residual

`experiments/notebooks/discovery_vacuum_hysteresis.py` tests whether a localized high-flux pulse leaves a spatially localized M_s residual. The selected report records peak center $M_s = 15.5843$ in `experiments/outputs/discovery_vacuum_hysteresis.h5`. This is a candidate simulation result and a future measurement protocol, not empirical evidence that physical vacuum memory has already been observed.

7 Numerical Safeguards

The main runtime now exposes bounded helpers in `runtime/diagnostics.py`. These guard toy diagnostics against non-finite values and keep failure modes visible. A non-finite result is not a discovery. It is a numerical or modeling failure unless the boundary condition is explicitly being studied as a breakdown surface.

8 Conclusion

The CDFD dynamics are useful only when their regime changes can be reproduced and falsified. The same mathematical language can describe recovery, overload, memory, and rupture, but the interpretation must always be tied to the specific domain mapping and to bounded numerical evidence.

References

- [1] Steve Bico Mujjabi, MD. *Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant*. CDFD Part I Paper I, preprint, 2026.
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